NMTA Team 5 — Project Plan (revised)

Q15 · Association rules · Lumen Beauty · Revised after instructor feedback

Team number	Team 5
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Question ID	Q15: What should the “you may also like” shelf show, and to whom?
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1. Problem & what it is worth

Which products should Lumen Beauty display on the post-purchase “you may also like” shelf, and should the shelf differ for first-time / light buyers vs heavy buyers? Stakeholder: brand founder and e-commerce merchandising. Worth: scenario ¥2.2–6.6k / month if 5–15% of ~127 monthly serum-only orders add a cream; sharper stake is whether a history-trained shelf would serve the heaviest buyers only.

2. Data

lb_order_lines + lb_users. Clean duplicates; treat empty stream_id on store orders as N/A. Build baskets two ways: by order_id and by user_id. Segment by n_orders (light 1–2, mid 3–5, heavy 6+), not high_value. 1.45 = distinct products per order, not units.

3. Method & assumptions

Apriori association rules (Week 3), pairs only. Report support, confidence, lift with co-purchase count and denominator. Thresholds from the shelf decision (orders ≥14 / 0.4%; histories 2%; categories 1%). Support/confidence are frequency filters, not significance tests. Theory: recommender bias + STP.

4. Expected results

We expect: (1) order baskets yield few, selective rules while histories flood with pairs; (2) Serum → Cream is the only strong order-level category rule and holds for light buyers and first orders; (3) history-rule evidence is dominated by heavy buyers; (4) top history pairs have same-order lift < 1. Expected pilot result: ≥5% of serum-only orders in the treatment half add a cream within six weeks.

5. Condition that would drop the action

Drop / do not roll out the new shelf if: (a) six-week treatment cream add-on rate < 2% and not above control; (b) light / first-time add-on rate is materially below heavy buyers; or (c) a re-mine on the control half shows Serum → Cream same-order lift ≤ 1. Also reject history-based “bought together” labelling for pairs with same-order lift ≤ 1.

6. Deliverables

Notebook (Restart & Run All), main report ≤12 pages, appendix (ethics + AI), 8-min pitch + Q&A. AI tools disclosed: Claude, Cursor.